

Open Source Reasoning Language Models

Fine-tune and Run DeepSeek, Kimi, Gemma, GPT-OSS and Other Open Reasoning Models on Your Hardware





Learning Objectives

By the end of this course, you will understand:

- Understand the differences between normal LLMs and reasoning LLMs
- Learn how to download, use and prompt reasoning models
- Understand the challenges of reasoning models (e.g. evaluation time)
- Get insights into the training process of reasoning models
 - Finetuning (GRPO)
 - Reinforcement learning
 - Advancements by DeepSeek, Kimi & Co





Agenda

- Introduction to Reasoning Models
- Differences to “normal” LLMs
- Try Reasoning Models
 - Open Router
 - Local Computer (with/out GPU)
- Techniques behind Reasoning Models
 - Reinforcement Learning
 - GRPO
 - Advances by DeepSeek



Introduction

- About me (Christian Winkler)
 - Programming for more than 40 years
 - PhD in physics, working as a professor at a university of applied science
- About the course
 - LLMs and the technology can be intimidating
 - Difficult to keep up-to-date
 - Let's try to clarify the myths
 - Hands on experience





Introduction to Reasoning Models



Count letters

How many „r“ are in strawberry?

- Famous question
- Should be in training set
- Many models fail

Problem

- Counting is easy, error detection is trivial
- But what happens with more complex problems?
- Idea: make models which challenge themselves
- Much better also for agentic tasks

How many "r" are in "strawberry"?



There are 2 "r"s in "strawberry".



Output of Llama-4-Scout-17B-16E-Instruct-Q4_K_M



Reasoning Approach

Expose process for insight generation

- Transparency
- Detect errors by understanding individual steps
- Get ideas to rephrase question

How many "r" are in "strawberry"?

📄 ✎ < 3/3 >

Thought Process

The word "strawberry" is spelled **S-T-R-A-W-B-E-R-R-Y**. Breaking it down:

1. **S**
2. **T**
3. **R** (first "r")
4. **A**
5. **W**
6. **B**
7. **E**
8. **R** (second "r")
9. **R** (third "r")
10. **Y**

There are **three "r"s** in the word.

Answer: 3



Output of Qwen_Qwen3-1.7B-Q4_K_M



Reasoning Process (1)

Okay, let's see. The user is asking how many "r"s are in the word "strawberry". Hmm, first I need to make sure I spell the word correctly. "Strawberry" is spelled S-T-R-A-W-B-E-R-R-Y. Wait, let me check again. S-T-R-A-W-B-E-R-R-Y. So breaking it down letter by letter.

Starting from the beginning: S, T, R, A, W, B, E, R, R, Y. So the letters are S, T, R, A, W, B, E, R, R, Y. Now I need to count the number of "r"s. Let's go through each letter.

First letter is S, not an r. Second is T, no. Third is R, that's one. Then A, W, B, E, then R again, that's two. Then another R, making it three. So total of three r's? Wait, but wait, the word is "strawberry". Let me double-check the spelling. Sometimes people might mispronounce it, but the standard spelling is S-T-R-A-W-B-E-R-R-Y. So the letters are R, R, and then another R? Wait, no. Let me count again. Let's write it out:

S (1), T (2), R (3), A (4), W (5), B (6), E (7), R (8), R (9), Y (10). So positions 3, 8, and 9. So that's three R's. But wait, sometimes when people say "strawberry", they might say it with two R's, but the actual spelling has three? Wait, let me check another way. The word is "strawberry", which is from the French "abricot", but the spelling in English is S-T-R-A-W-B-E-R-R-Y. So yes, two R's in the middle, but the third R is at the end? Wait, no. Wait, the last letter is Y. So the letters are S-T-R-A-W-B-E-R-R-Y. So that's R, R. So two R's? Wait, but the user wrote "strawberry" with two R's? Wait, maybe I'm confused. Let me check the spelling again.



Reasoning Process (2)

Wait, "strawberry" is spelled S-T-R-A-W-B-E-R-R-Y. So the letters are S, T, R, A, W, B, E, R, R, Y. So there are two R's in the middle, but the third R is... no, wait, the third R is the second R. Wait, the letters are:

1. S
2. T
3. R
4. A
5. W
6. B
7. E
8. R
9. R
10. Y

So positions 3, 8, and 9. So three R's? Wait, but in the middle, after the "w", there's "b", "e", then "r", "r". So two R's? Wait, no, the third letter is R, then the next two are R again. So that's three R's? Wait, no, position 3 is the first R, then position 8 and 9 are the second and third R's? Wait, that's three R's. But sometimes people might think that "strawberry" has only two R's. Let me check an example.



Reasoning Process (3)

If I type "strawberry" into a search engine, it would show the spelling as S-T-R-A-W-B-E-R-R-Y. So yes, that's two R's in the middle, but the third R is actually the second one. Wait, no, the letters are S, T, R, A, W, B, E, R, R, Y. So there are two R's in the word. Wait, position 3 is the first R, then positions 8 and 9 are the second and third R's? No, position 8 is the second R, and position 9 is the third R? Wait, but the word is "strawberry" which is 10 letters. Let me count again:

1. S
2. T
3. R
4. A
5. W
6. B
7. E
8. R
9. R
10. Y



Reasoning Process (4)

So positions 3, 8, 9. That's three R's. Wait, but the word is "strawberry", which is 10 letters. Wait, but "strawberry" is spelled with two R's. Wait, maybe I'm making a mistake here. Let me think. The word "strawberry" is pronounced "straw-berry", so the spelling would be S-T-R-A-W-B-E-R-R-Y. So the R's are in the third and eighth positions, and the ninth is another R? Wait, that's three R's. But I thought that "strawberry" only has two R's. Maybe there's confusion with the word "strawberry" vs. "strawberry" with a different spelling. Wait, maybe the user is confused because they think "strawberry" has two R's, but actually it has three. Wait, but in the standard English spelling, "strawberry" is spelled with two R's. Let me check a dictionary. But since I can't access external sources, I need to rely on my memory.

In the word "strawberry", the R's are in the third and eighth positions. Wait, no, let's see:

Breaking it down:

S-T-R-A-W-B-E-R-R-Y

So the R is the third letter, then after the W, it's B, E, R, R. So that's two R's. Wait, so positions 3, 8, and 9? Wait, position 3 is the first R, then after B, E, then R, then another R. So that's three R's? Wait, but the word is S-T-R-A-W-B-E-R-R-Y. So after the first R, which is position 3, then A, W, B, E, then R, R. So positions 3, 8, 9. So that's three R's. But that would make the word have three R's. However, when I think of "strawberry", the R's are at the end, but the spelling is S-T-R-A-W-B-E-R-R-Y. So the third R is after the W, which is at position 8, and the second R is at position 9.



Reasoning Process (5)

Wait, maybe I'm miscounting. Let me write it out:

1. S
2. T
3. R
4. A
5. W
6. B
7. E
8. R
9. R
10. Y

Yes, positions 3, 8, and 9. So that's three R's. Therefore, the answer would be three. But I'm a bit confused because sometimes people might think it's two. But according to the spelling, it's three. So the answer should be three.



Reasoning Process (final answer)

The word "strawberry" is spelled **S-T-R-A-W-B-E-R-R-Y**. Breaking it down:

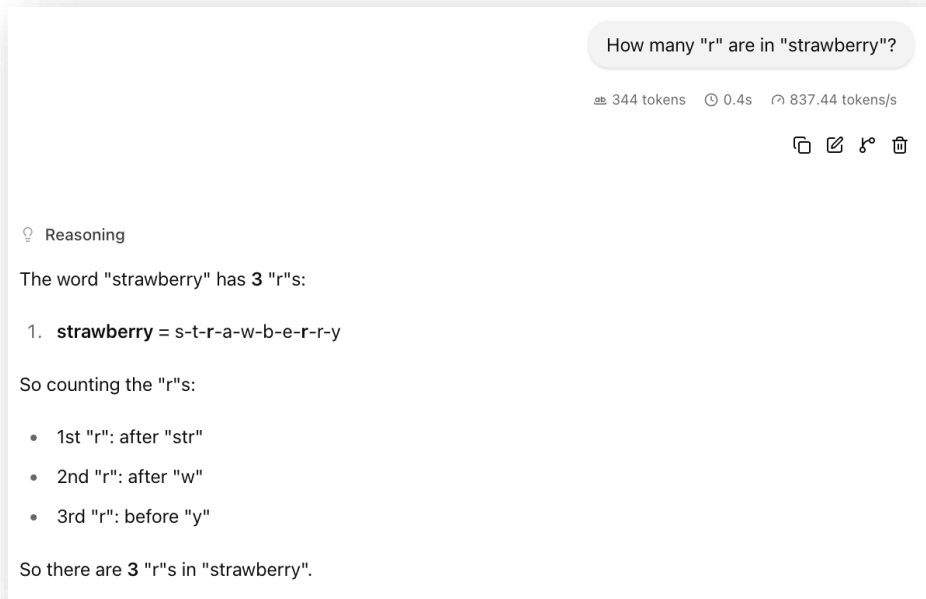
1. S
2. T
3. R (first "r")
4. A
5. W
6. B
7. E
8. R (second "r")
9. R (third "r")
10. Y

There are **three "r"s** in the word.

Answer: 3



New Model: Qwen3.5-0.8B: Optimized Thinking



The user is asking about the word "strawberry" and wants to know how many "r"s are in it. This is a simple question that doesn't require me to use any of the available tools.

Let me count the "r"s in "strawberry":
s-t-r-a-w-b-e-r-r-y

There are three "r"s in the word:

The 3rd letter is "r" (strawberry)

The 5th letter is "r" (strawberry)

The 8th letter is "r" (strawberry)

So there are 3 "r"s in "strawberry".



How does it work?

Models are *finetuned*

- Special reasoning dataset
- Happens after Supervised Finetune
- Originally with o1 mini (preview)

Finetuning process has been heavily optimized

- Many new ideas
- Reasoning finetuning is now also possible with moderate hardware
- Datasets are available

> What is the sum of 13 and 25?

<think>

Okay, let's see. The question is asking for the sum of 13 and 25. Hmm, sum means them together, right? So I need to add 13 and 25. Let me think about how to do this.

First, maybe I can break it down. 13 plus 25. Let me add the ones place first. 13 has 3 ones, and 25 has 5 ones. So 3 plus 5 is 8. Then the tens place: 1 ten from 13 and 2 tens from 25. So 1 + 2 is 3. So combining those, it's 38? Wait, is that right?

...

</think>

To find the sum of 13 and 25, we simply add the two numbers together:

$$13 + 25 = 38$$

This can be verified by breaking down the numbers:

- Add the ones place: $3 + 5 = 8$
- Add the tens place: $1 + 2 = 3$
- Combine the results: 38

Thus, the final answer is: 38



Why are Reasoning Models Popular?

MODEL	OVERALL ▼	REASONING	CODING	AGENTIC CODING	MATHEMATICS	DATA ANALYSIS	LANGUAGE	INSTRUCTION FOLLOWING	COST PER SUCCESSFUL TASK
• Claude Fable 5.1 Max Effort	83.4	91.7	86.4	66.1	97.0	80.3	89.5	73.0	\$1.212
• Claude Fable 5 Max Effort	83.0	89.7	86.0	62.2	96.0	80.5	90.7	75.8	\$1.439
• GPT-6 Astra Max Effort	82.2	92.7	80.4	57.3	96.8	83.0	89.4	75.6	\$0.736
• Muse Spark 1.3 xHigh Effort	81.6	89.7	81.1	64.1	95.9	79.6	82.8	78.0	\$0.219
• DeepSeek V4.1 Flash Max Effort open	81.1	86.7	80.0	77.3	93.3	79.3	81.2	70.0	\$0.029
• GPT-5.6 Sol Max Effort	81.0	91.7	83.9	56.2	96.2	79.8	87.7	71.8	\$0.515
• GPT-5.5 Thinking xHigh Effort	80.2	89.7	82.1	54.0	95.9	81.6	87.4	70.7	\$0.435
• Claude 5 Opus Thinking Max Effort	80.1	91.2	81.4	65.2	95.7	74.6	88.7	63.8	\$0.699
• Kimi K3 open	79.2	90.7	81.4	62.2	84.4	78.7	85.5	71.4	\$0.348
• Gemini 3.7 Flash High	78.8	87.8	78.9	58.3	93.5	68.0	85.5	79.9	\$0.157
• Qwen 3.8 Max open	78.5	88.2	72.9	64.6	91.3	78.4	79.7	74.1	\$0.275
• Grok 4.6	78.0	90.5	76.8	57.0	92.6	73.9	83.7	71.9	\$0.207
• GPT-5.4 Thinking xHigh Effort	78.0	88.1	77.5	53.8	94.1	79.3	82.6	70.2	\$0.387
• Muse Spark 1.2 xHigh Effort	78.0	90.0	77.5	57.6	91.2	76.5	78.6	74.3	\$0.375
• GPT-5.6 Terra Max Effort	77.9	90.6	78.2	54.9	94.9	79.3	82.9	64.6	\$0.352
• DeepSeek V4 Pro 0813 open	77.4	85.8	77.2	54.9	95.1	79.2	82.1	67.7	\$0.044
• Gemini 3.1 Pro Preview High	77.0	84.0	76.5	44.1	91.0	78.5	85.4	79.1	\$0.286



Why are Reasoning Models Popular?

Rank	Model	Net Improvement	Confirmed Success	Praise vs Complaint	Steerability
1 1 → 2	Claude Fable 5.1 (Max) Anthropic · Proprietary	▲ 13.71% ±1.72%	▲ 19.83% ±2.75%	▲ 31.83% ±6.86%	▲ 3.88% ±3.61%
2 1 → 6	GPT 6 Astra (Max) OpenAI · Proprietary	▲ 11.54% ±2.10%	▲ 17.70% ±3.26%	▲ 32.79% ±8.37%	▼ 0.47% ±4.67%
3 2 → 6	Claude Opus 5 (High) Anthropic · Proprietary	▲ 10.25% ±1.41%	▲ 9.24% ±2.94%	▲ 18.66% ±5.22%	▲ 11.04% ±2.75%
4 2 → 6	Claude Opus 5 (Max) Anthropic · Proprietary	▲ 10.16% ±1.55%	▲ 12.41% ±3.04%	▲ 18.15% ±5.77%	▲ 6.83% ±3.05%
5 2 → 8	Claude Fable 5 (High) Anthropic · Proprietary	▲ 8.81% ±1.25%	▲ 5.97% ±2.66%	▲ 18.07% ±4.46%	▲ 10.60% ±2.28%
6 2 → 8	Claude Opus 4.8 (High) Anthropic · Proprietary	▲ 8.19% ±1.27%	▲ 6.28% ±2.47%	▲ 16.23% ±4.59%	▲ 11.06% ±2.22%
7 5 → 13	GPT 5.6 Sol (xHigh) OpenAI · Proprietary	▲ 7.10% ±1.28%	▲ 3.74% ±2.61%	▲ 20.48% ±4.62%	▲ 6.62% ±2.41%
8 7 → 13	Kimi K3 (Max) Moonshot · Kimi K3 license	▲ 6.22% ±0.62%	▲ 11.91% ±1.28%	▲ 11.79% ±2.18%	▲ 1.99% ±1.28%
9 5 → 16	Claude Sonnet 5 (High) Anthropic · Proprietary	▲ 5.97% ±1.62%	▲ 2.88% ±3.34%	▲ 11.03% ±5.77%	▲ 8.39% ±3.38%
10 7 → 17	GPT 5.5 (xHigh) OpenAI · Proprietary	▲ 5.03% ±0.92%	▼ 0.61% ±2.02%	▲ 9.14% ±3.18%	▲ 6.61% ±1.77%
11 7 → 17	Hy4 preview Tencent · Apache 2.0	▲ 5.01% ±1.02%	▲ 9.85% ±1.98%	▲ 8.76% ±3.77%	▼ 1.23% ±2.16%
12 7 → 19	Deepseek V4.1 Flash (Max) DeepSeek · MIT	▲ 4.88% ±1.33%	▲ 13.35% ±2.49%	▲ 4.76% ±4.56%	▼ 1.63% ±3.40%
13 7 → 21	Gemini 3.8 Flash (High) Google · Proprietary	▲ 4.71% ±1.91%	▲ 9.30% ±6.70%	▲ 13.34% ±6.70%	▼ 1.22% ±4.83%
14 9 → 18	GLM 5.2 (Max) Z.ai · MIT · SiliconFlow	▲ 4.37% ±0.69%	▲ 4.90% ±1.50%	▲ 9.76% ±2.42%	▲ 4.88% ±1.31%
15 9 → 20	Muse Spark 1.3 (Max) Meta · Proprietary	▲ 4.20% ±0.86%	▲ 10.31% ±1.71%	▲ 2.94% ±2.94%	▲ 1.47% ±1.94%

Source: arena.ai

Models 39,405

Filter by name

Base only

Inference Available

Sort: Trending



deepseek-ai/DeepSeek-V4.1-Flash

Image-Text-to-Text · 763B · Updated 7 days ago · 391k · 2.94k

ISTA-DASLab/Qwen3.8-27B-GSQ-RC0-GGUF

Image-Text-to-Text · 27B · Updated 15 days ago · 1.03M · 1.23k

ukisai/Swift-Qwen3.8-27b

Image-Text-to-Text · 28B · Updated about 22 hours ago · 3.22k · 358

dealignai/DeepSeek-V4.1-Flash-UNCENSORED-FP8

Image-Text-to-Text · 763B · Updated 6 days ago · 32k · 255

ukisai/Swift-Qwen3.8-27B-GGUF

Image-Text-to-Text · 27B · Updated about 22 hours ago · 72.9k · 210

TaichuAI/ZDTaichu5.0-9B

Image-Text-to-Text · 10B · Updated 2 days ago · 476 · 164

HauhuaiCS/Qwen3.8-27B-Uncensored-HauhuaiCS-Aggressive...

Image-Text-to-Text · 2B · Updated about 1 month ago · 2.35M · 1.25k

huihui-ai/Huihui-Qwen3.8-27B-abliterated-GGUF

Image-Text-to-Text · 27B · Updated 9 days ago · 2.86M · 768

ISTA-DASLab/Qwen3.8-Flash-Next-GSQ-RC0-GGUF

Image-Text-to-Text · 177B · Updated 2 days ago · 13.5k · 100

Qwen/Qwen3.8-27B

Image-Text-to-Text · 28B · Updated Aug 14 · 7.46M · 15.5k

DavidAU/Qwen3.8-27B-TURBO-Fable-Cold-Fusion-735-882...

Image-Text-to-Text · 27B · Updated about 13 hours ago · 1.12M · 826

Qwen/Qwen3.8-Flash-Next

Image-Text-to-Text · 180B · Updated 21 days ago · 706k · 5.34k

Agnes-AI/Agnes-3.0-Flash

Image-Text-to-Text · 33B · Updated 1 day ago · 1.22k · 208

zai-org/GLM-5.3-Flash

Image-Text-to-Text · 321B · Updated 10 days ago · 2.45M · 2.41k

DavidAU/Qwen3.8-27B-TWIN-TURBO-Fable-Cold-Fusion-70...

Image-Text-to-Text · 27B · Updated about 5 hours ago · 134k · 151

orcarouter/Qwen3.8-27B-Uncensored-GGUF

Image-Text-to-Text · 27B · Updated 21 days ago · 333k · 1.03k

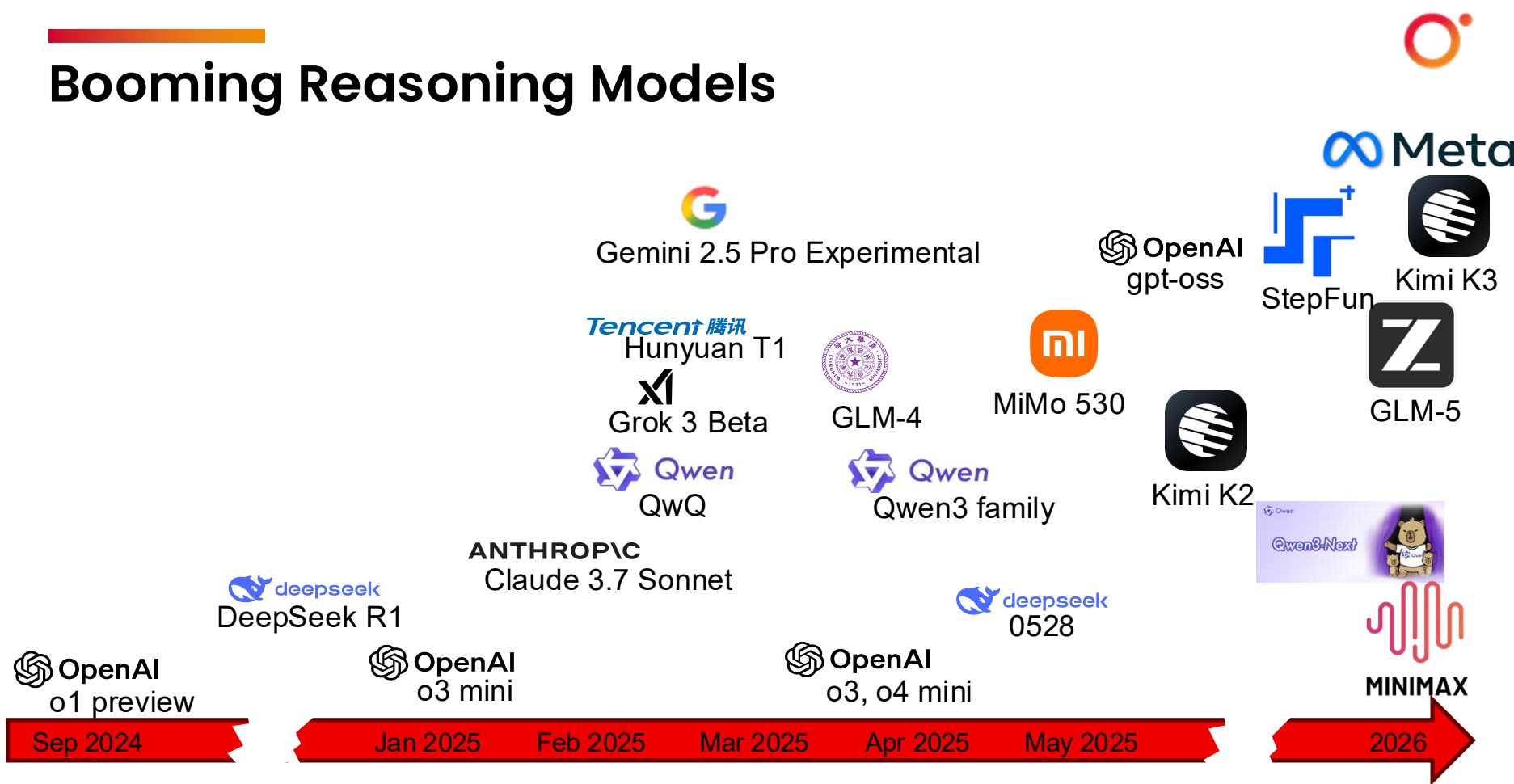
unsloth/Qwen3.8-Flash-Next-GGUF

Image-Text-to-Text · 177B · Updated 15 days ago · 1.42M · 980

cyjin-y1/Qwen3.8-27B-Uncensored-Cyber-agentic-imatr...

Image-Text-to-Text · 3.39M · Updated 28 days ago · 14.8k · 114

Booming Reasoning Models



Sudden Interest?

First reasoning model from OpenAI (o1 preview)

- Expensive and an experiment (preview)

Booster: Publication of DeepSeek R1

- Big surprise – big panic
- New ideas, *everything Open Source* (except for training data!)

Very surprising?

- Not really, DeepSeek Coder, DeepSeek V3 were there before

AI knowledge coming from nothing?

- No, AI was used for Hedge Fund (High-Flyer)
- Earned lots of money

Strategic AI competences for China





Q&A





Try Reasoning Models



Different Alternatives

Local Installation with GPU

- Hugging Face [transformers](#)
- [vLLM](#)
- [llama.cpp](#)

Local Installation without GPU

- [llama.cpp](#)
- [mlx](#) (Apple Silicon)

Service Provided by an API

- [DeepSeek](#)
- [Qwen](#)
- [Kimi](#)
- [MiniMax](#)
- [GLM](#)

Use [OpenRouter](#)

- Free services available
- Less (or no) tracking

DeepSeek-like models

- DeepSeek
 - Too large (except for V4 Flash)
- Kimi K3
 - Arguably most powerful open model
 - From Moonshot.ai
 - Copied DeepSeek architecture
 - 2.8T parameters (!)



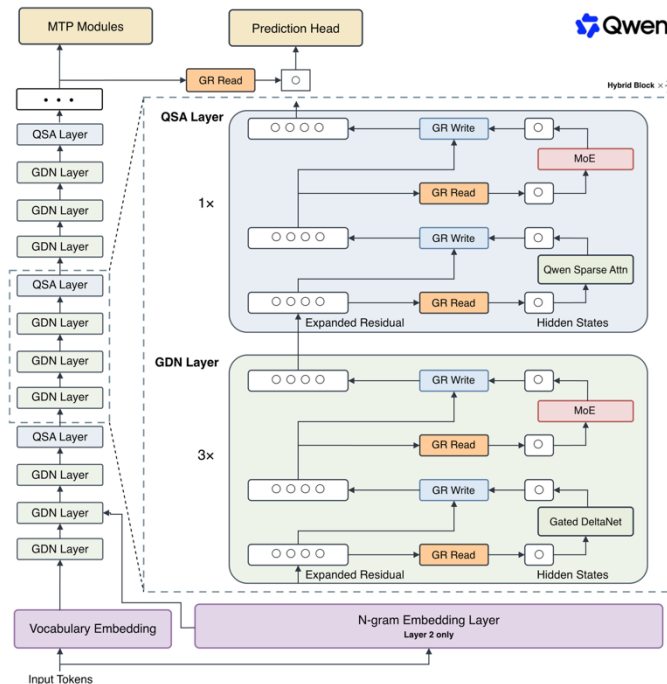
Qwen3.{5,6,8} models



Qwen



- Model family
 - Different sizes
 - Different architectures
- Small models
 - Suitable for finetuning
- Medium models
 - Quite powerful
 - Working on GPUs and CPUs
- Big models
 - Mainly for cloud
 - New architecture: Qwen3.8-Flash-Next (very powerful)



Other models

- Gemma 4 from Google
- GPT-OSS from Open AI (**outdated**)
- GLM from Z.ai
 - Very powerful (GLM 5.3), Flash models not too large
- MiniMax M3
 - Powerful and small
- MIMO V2.5
 - Different sizes upto 1T
- Many others, like very small Nanbeige

Gemma

OpenAI



MINIMAX



xiaomi



Live Demo Jupyter with transformers, llama.cpp, mlx





Q&A





Techniques behind Reasoning Models



Instruction following

- LLMs are trained to predict the *next word*
- Very good for completing text
- But how can they answer questions?
- Basic idea: instruction following
 - Finetune models for question/answer pairs
 - Use completion capabilities
 - Add a *super structure*

```
</system/>
You are a helpful travel assistant.</end/>
</user/>
I am going to Paris, what should I see?</end/>
</assistant/>
<think>...</think>
Paris, the capital of France, is known for its stunning
architecture, art museums, historical landmarks, and
romantic atmosphere. Here are some of the top attractions to
see in Paris:

1. The Eiffel Tower: The iconic Eiffel Tower is one of the
most recognizable landmarks in the world and offers
breathtaking views of the city.
2. The Louvre Museum: The Louvre is one of the world's
largest and most famous museums, housing an impressive
collection of art and artifacts, including the Mona Lisa.
3. Notre-Dame Cathedral: This beautiful cathedral is one of
the most famous landmarks in Paris and is known for its
Gothic architecture and stunning stained glass windows.

These are just a few of the many attractions that Paris has
to offer. With so much to see and do, it's no wonder that
Paris is one of the most popular tourist destinations in the
world.</end/>
```

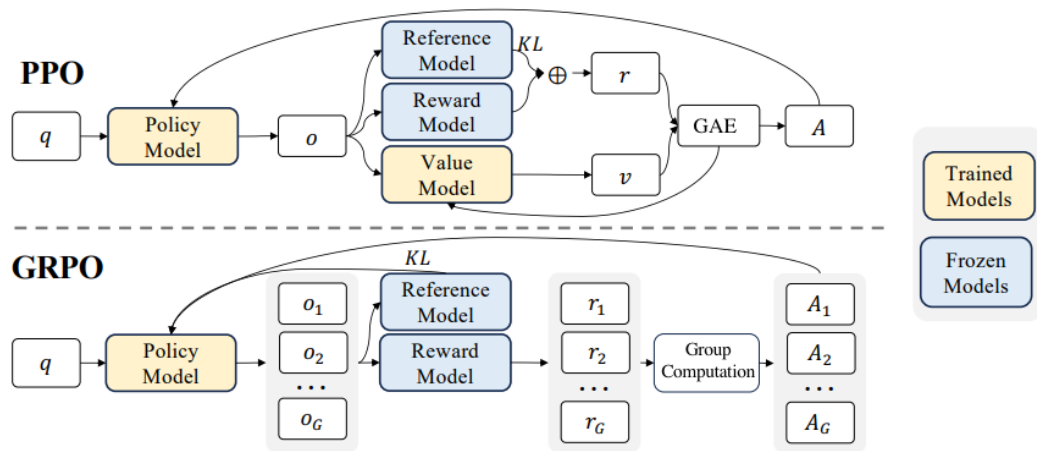
Innovations by DeepSeek (1)

Reinforcement Learning

- Already used in older models (RLHF in SFT stage)
- Now in multiple stages

New method for RL: GRPO

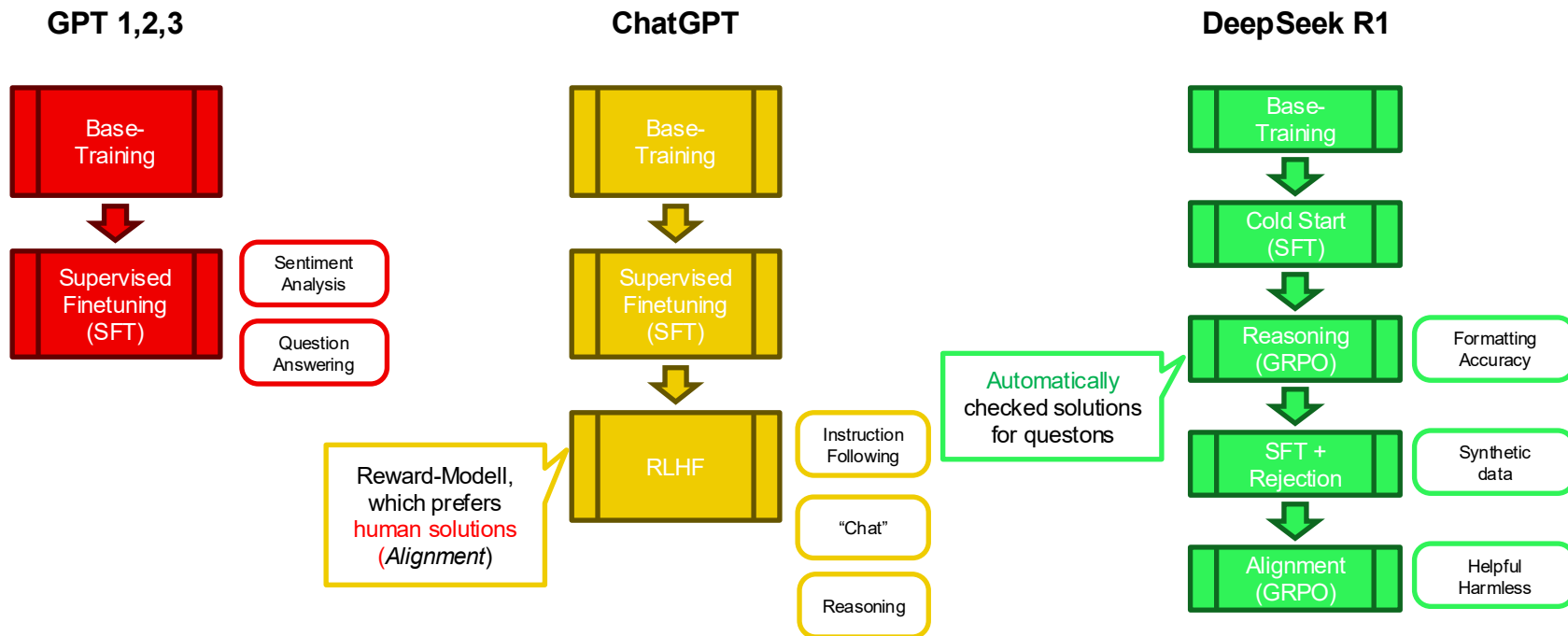
- Group Relative Policy Optimization
- Avoids value model
- Checks different alternatives
- Builds ranking



Source: <https://arxiv.org/pdf/2402.03300>



Focus: New Finetuning Process (GRPO, RL)





Innovations by DeepSeek (2)

Multi-head Latent Attention (MLA)

Multi-Token Prediction

Low-Level optimization (PTX)

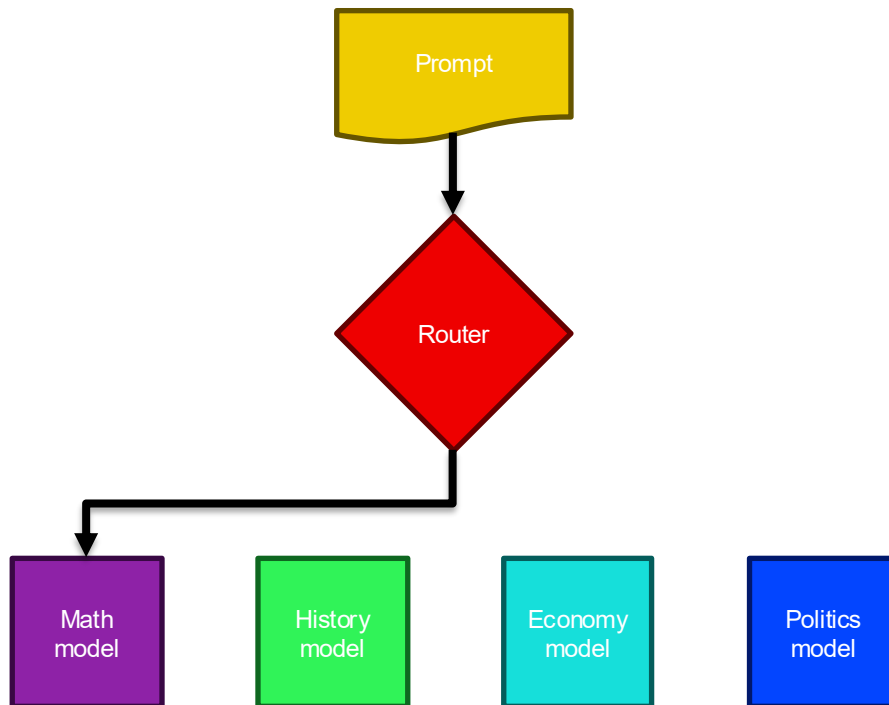
- Hardware export not allowed
- Only few(er) GPUs available
- Training more efficient and cheaper
- Much cheaper inference
(\$ per token)

Weights as FP8

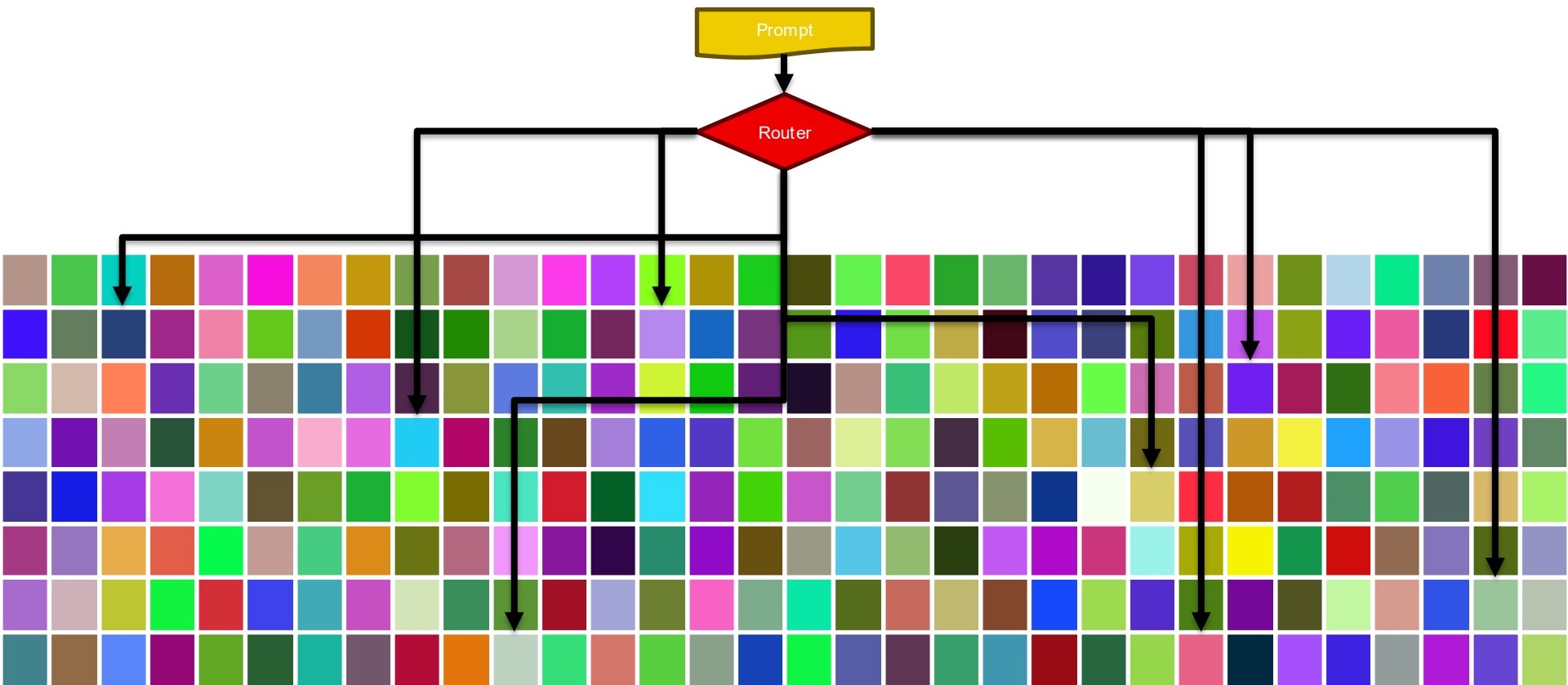
- Saves 50% VRAM and doubles inference speed (bandwidth)
- More sophisticated training process (once)

Architecture: Sparse Mixture of Experts with Auxiliary-Loss-Free Load Balancing

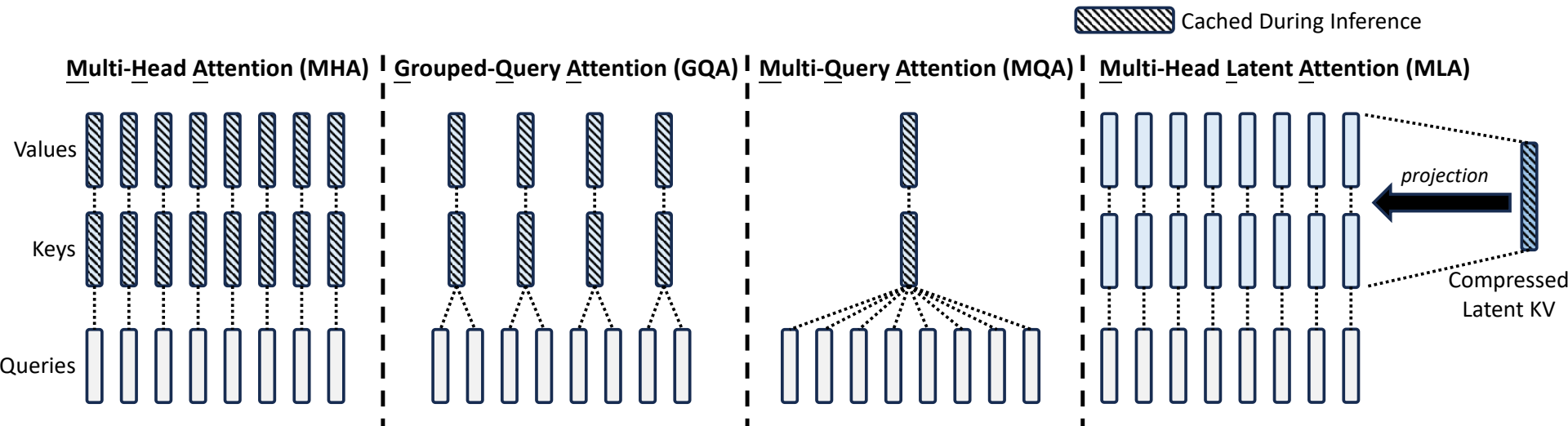
Focus: Sparse Mixture of Experts (SMoE, traditional)



Focus: SMoE (DeepSeek 671B-37A)



Focus: Multi-head latent attention

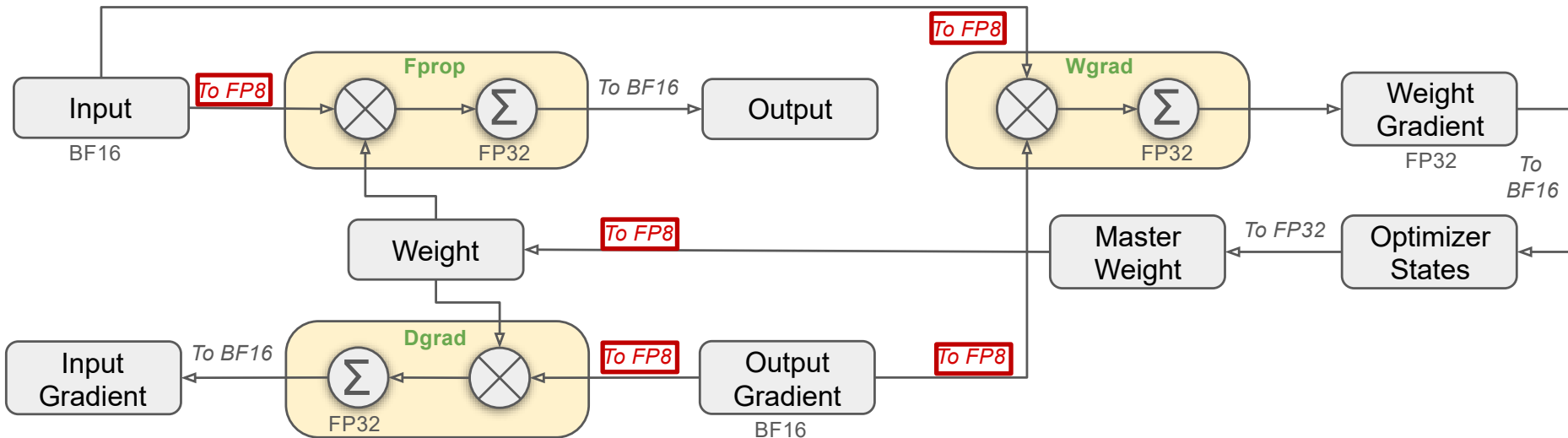


Source: <https://arxiv.org/pdf/2405.04434>

Many new attention mechanisms like hybrid attention, compressed sparse attention (2) etc.



Focus: FP8 training (mixed precision)



Source: <https://arxiv.org/abs/2412.19437>

Some models are now trained with NVFP4 or MXFP4 to save even more space!
But the general procedure was first applied by DeepSeek.

Is DeepSeek an Open Model?



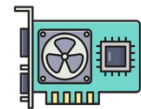
Much more open compared to OpenAI

- Architecture and methods published
- Weights publically available
- Can run on own hardware

Much information missing

- Training data (objectivity, Open AI → Google?)
- Heavily aligned for China-critical topics

Same is true for almost all other big models, except Nvidia Nemotron

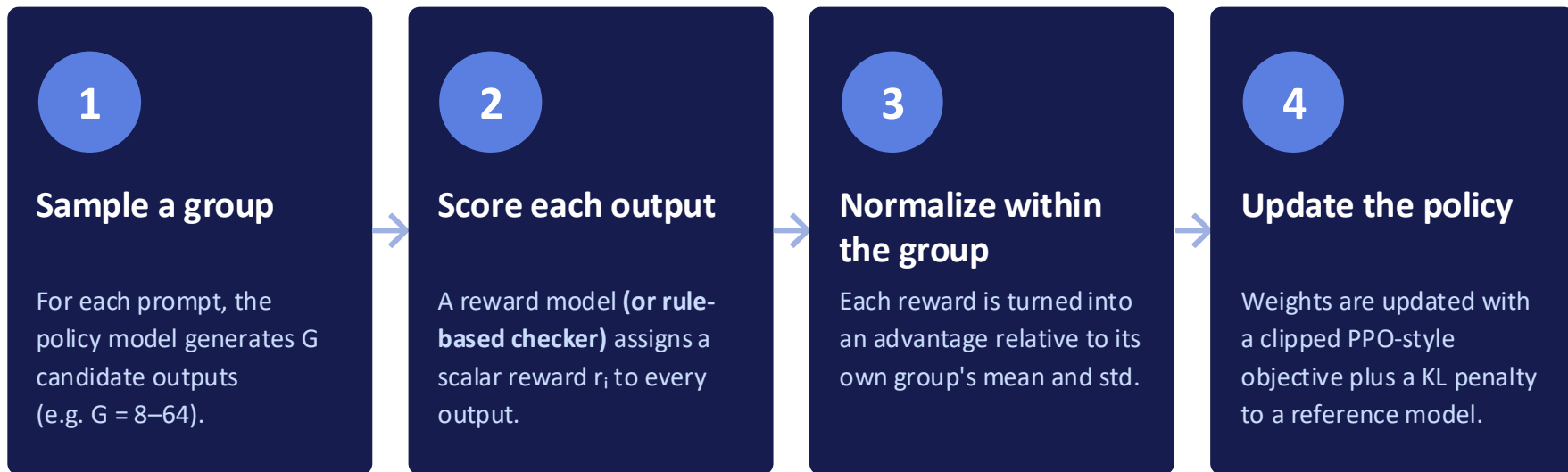




Training (i.e. Finetuning) Reasoning Models



GRPO training procedure



Reinforcement-learning finetuning method that scores a group of sampled responses against each other
no separate critic network required



GRPO training data

Training data contains questions and answers

- Answers for GRPO are often separated into solution path and final answer
- **Rewards** given for
 - Correct form (path + answer)
 - Correct path
 - Correct answer

Take a detailed look at one example:

What is the coefficient of x^2y^6 in the expansion of $\left(\frac{3}{5}x - \frac{y}{2}\right)^8$? Express your answer as a common fraction.

To determine the coefficient of (x^2y^6) in the expansion of $\left(\left(\frac{3}{5}x - \frac{y}{2}\right)^8\right)$, we can use the binomial theorem.

The binomial theorem states:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

In this case, $(a = \frac{3}{5}x)$, $(b = -\frac{y}{2})$, and $(n = 8)$.

We are interested in the term that contains (x^2y^6) . In the general term of the binomial expansion:

$$\binom{8}{k} \left(\frac{3}{5}x\right)^{8-k} \left(-\frac{y}{2}\right)^k$$

To get (x^2) , we need $(8 - k = 2)$, thus $(k = 6)$.

Substituting $(k = 6)$ into the expression:

$$\binom{8}{6} \left(\frac{3}{5}x\right)^{8-6} \left(-\frac{y}{2}\right)^6 = \binom{8}{6} \left(\frac{3}{5}x\right)^2 \left(-\frac{y}{2}\right)^6$$

Now, we will compute each part of this expression.

1. Calculate the binomial coefficient $\left(\binom{8}{6}\right)$.
2. Compute $\left(\left(\frac{3}{5}\right)^2\right)$.
3. Compute $\left(\left(-\frac{y}{2}\right)^6\right)$.
4. Combine everything together to get the coefficient of (x^2y^6) .

Let's compute these in Python.

```
from math import comb

# Given values
n = 8
k = 6

# Calculate the binomial coefficient
binom_coeff = comb(n, k)

# Compute  $(3/5)^2$ 
a_term = (3/5)**2

# Compute  $(-1/2)^6$ 
b_term = (-1/2)**6

# Combine terms to get the coefficient of  $x^2y^6$ 
coefficient = binom_coeff * a_term * b_term
print(coefficient)

0.1575
```

The coefficient of (x^2y^6) in the expansion of $(\left(\frac{3}{5}x - \frac{y}{2}\right)^8)$ is (0.1575). To express this as a common fraction, we recognize that:

$$0.1575 = \frac{1575}{10000} = \frac{63}{400}$$

Thus, the coefficient can be expressed as:


$$\frac{63}{400}$$

GRPO scores

Reward function

- Correct answer can be found in the **box**
- Python programs can be checked if they are syntactically correct
- Different words (like binomial coefficient) can contribute to the score
- etc.

Reward model

- Sometimes, an existing model is used to check the result and assign scores

GRPO vs. RLVR

GRPO is the optimization algorithm (group-normalized vs. PPO critic's).

RLVR is the reward design.

In practice, this is almost always combined!



Live Demo

Train your own reasoning model





Q&A





Summary & discussion





Group Discussion

- Any questions left?
- What would you like to achieve with this technology?

Summary & Outlook

Deep Research helps science

- Basis: reasoning (combined with RAG)
- Existing applications like Manus



New interesting scenarios are explored

- Find flaws in scientific articles
- Generate new ideas

Methods and technologies from DeepSeek

- RL, GRPO, MoE standard in all new models
- New: Reinforcement Learning with Verifiable Rewards

Size of models decreasing

- Qwen3.5 has reasoning with <1B parameters

Models are getting faster

- MoE allows fast inference for large models
- Important because of long *thinking* process



→ **Democratizing reasoning models**

→ **New (local) applications possible**



Review Course Outcomes

By the end of this course, you will understand:

- Understand the differences between normal LLMs and reasoning LLMs
- Learn how to download, use and prompt reasoning models
- Understand the challenges of reasoning models (e.g. evaluation time)
- Get insights into the training process of reasoning models
 - Finetuning (GRPO)
 - Reinforcement learning
 - Advancements by DeepSeek



Resources



GitHub repository

- <https://github.com/datanizing/oreilly-open-source-reasoning>

General Discussion

- <https://www.reddit.com/r/LocalLLaMA/>
- Mixed audience, be careful!

Used programs and libraries

DeepSeek models and architecture

- <https://huggingface.co/deepseek-ai>
- <https://arxiv.org/pdf/2402.03300>
- <https://arxiv.org/pdf/2501.12948>
- <https://arxiv.org/pdf/2505.09343>

Qwen models and architecture

- <https://huggingface.co/Qwen>
- <https://qwenlm.github.io/blog/qwen3/>

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